

Library Information System

Software Engineering Lab project

Team

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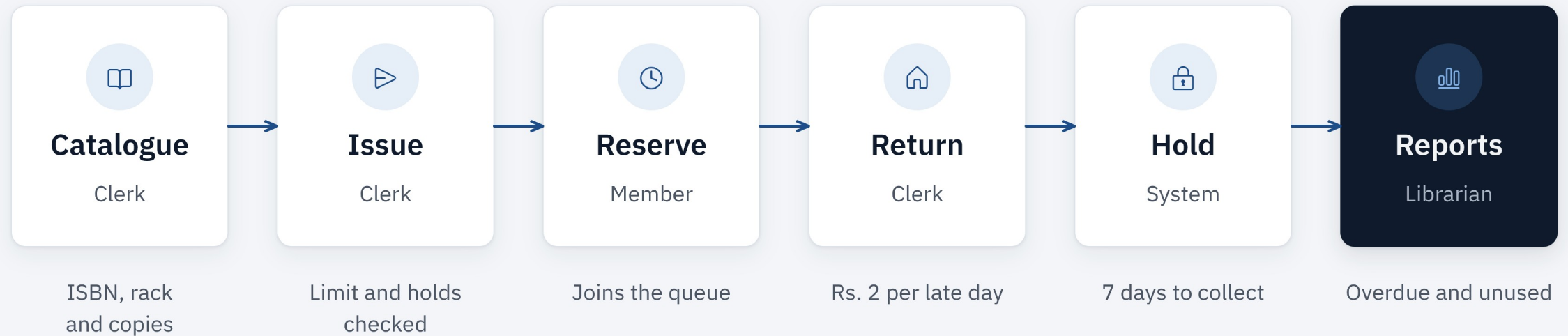


A paper register can't answer three daily questions

- 01 Who has this copy right now?
- 02 When is it due, and what is the fine?
- 03 Who asked for it first?

 LIS answers each one from a single database.

How a book moves through the system



Three roles, three views



Librarian

Manages members
Prints overdue reminders
Checks unused books



Clerk

Adds and removes books
Issues books
Takes returns and fines



Member

Searches the catalogue
Reserves books

Functional requirements

ID	Requirement	Actor
FR1	Sign in by role or member code	All
FR2	Add a book with ISBN, rack and copies	Clerk
FR3	Remove a book with no copy on loan	Clerk
FR4	Search by ISBN, title or author	All
FR5	Register and remove members	Librarian
FR6	Issue a book within the category limit	Clerk
FR7	Return a book and charge any fine	Clerk
FR8	Reserve a book with no free copy	Member
FR9	Print overdue reminders	Librarian
FR10	List books unused for five years	Librarian

Business rules

Loan limits by category

Category	Books	Days
Undergraduate	2	30
Postgraduate	4	30
Research scholar	6	90
Faculty	10	180

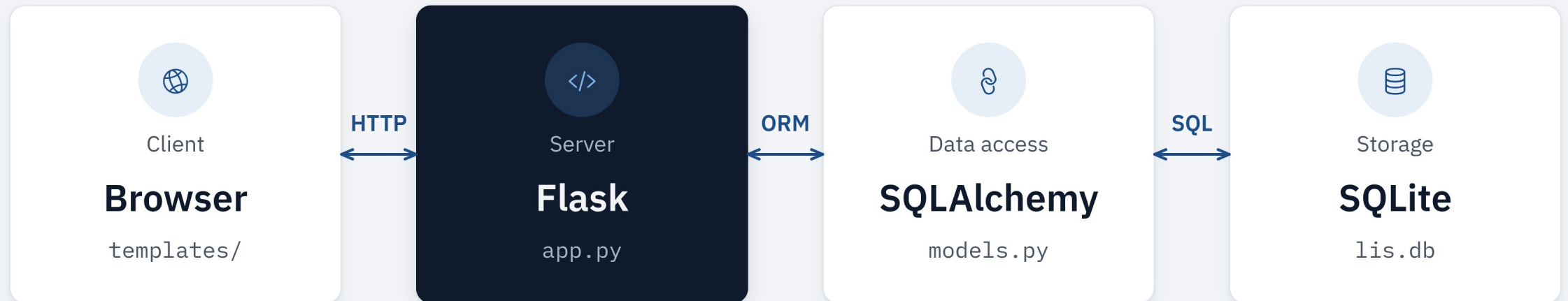
Other rules

Rule	Value
Late fine	Rs. 2 / day
Reservation hold	7 days
Unused report	5 years

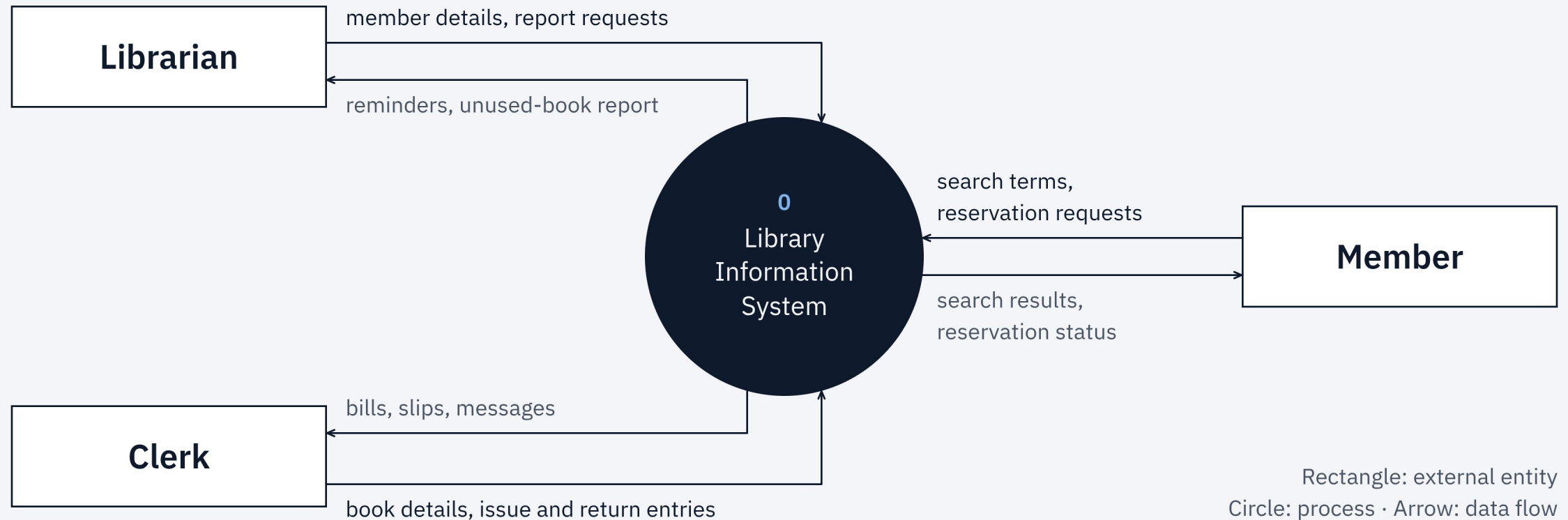
Tech stack

Technology	Version	Used for
Python	3.11	Language
Flask	3.0.0	Web server and routing
Flask-SQLAlchemy	3.1.1	Database models (ORM)
SQLite	Built in	Database file
Jinja2	With Flask	HTML templates
Bootstrap	5.3.0	Layout and forms
pytest	7.4.3	Unit tests
Git and GitHub	—	Version control

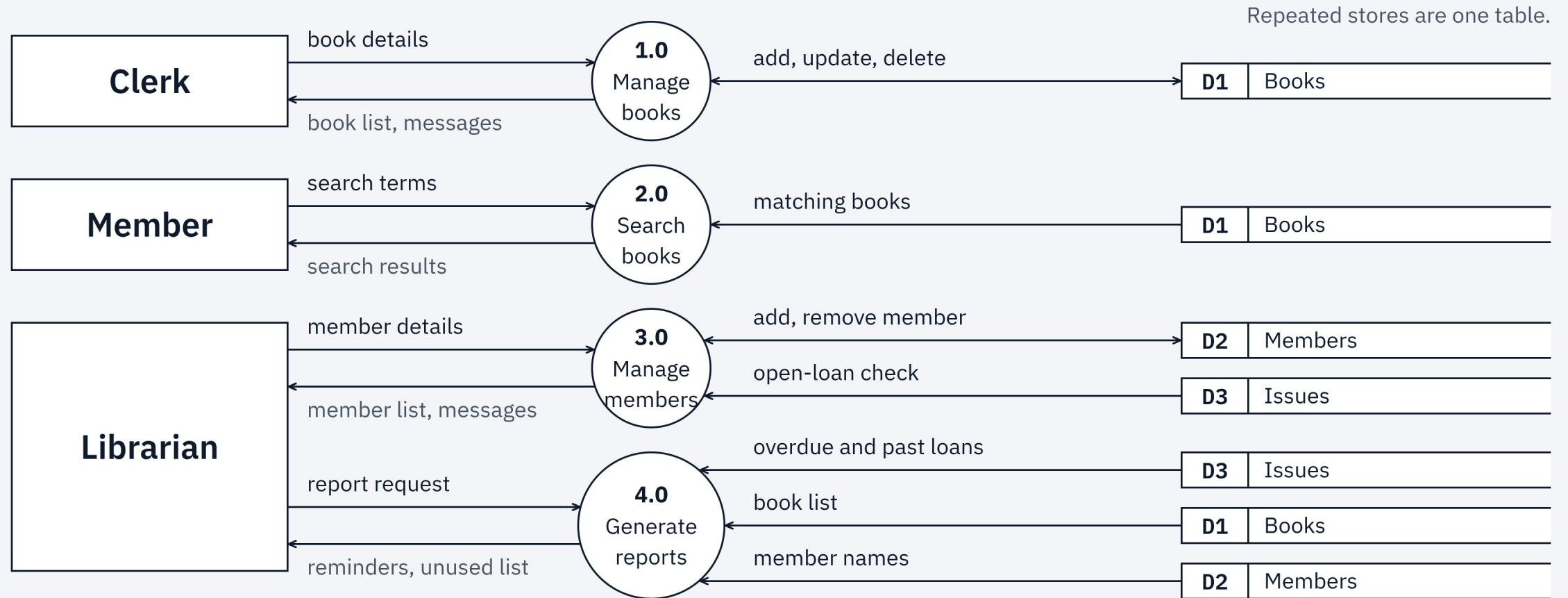
System architecture



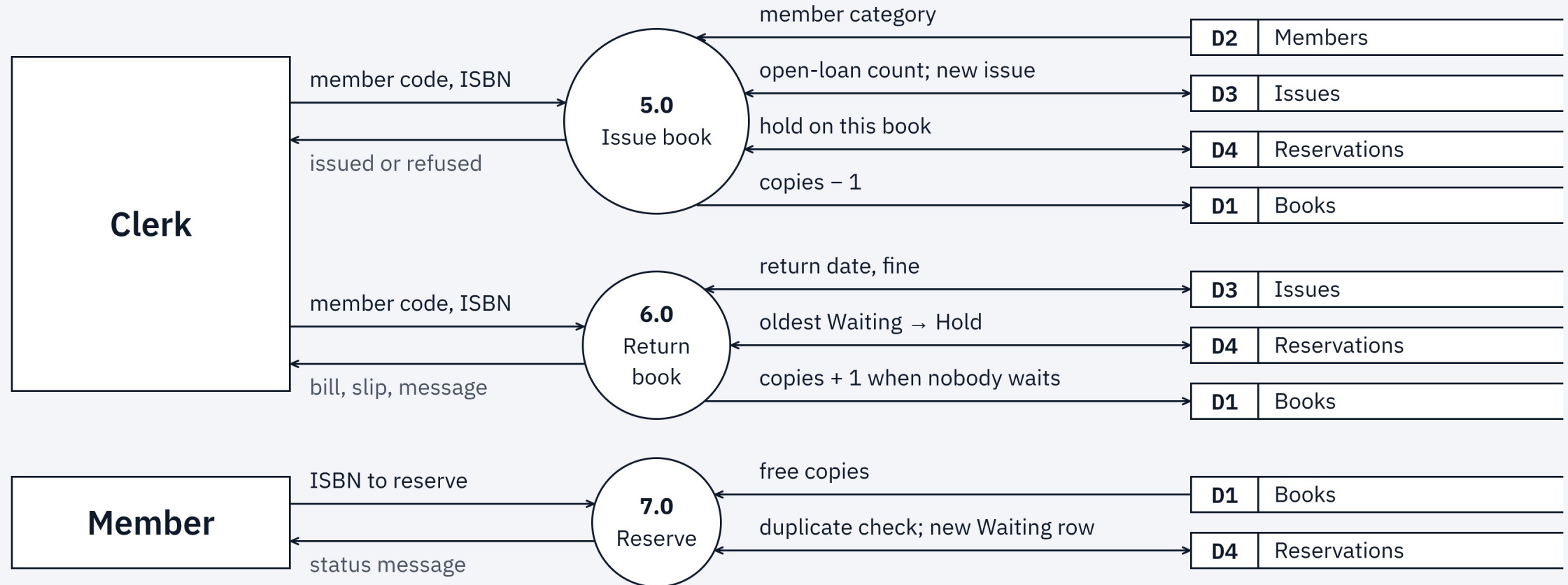
DFD level 0: context diagram



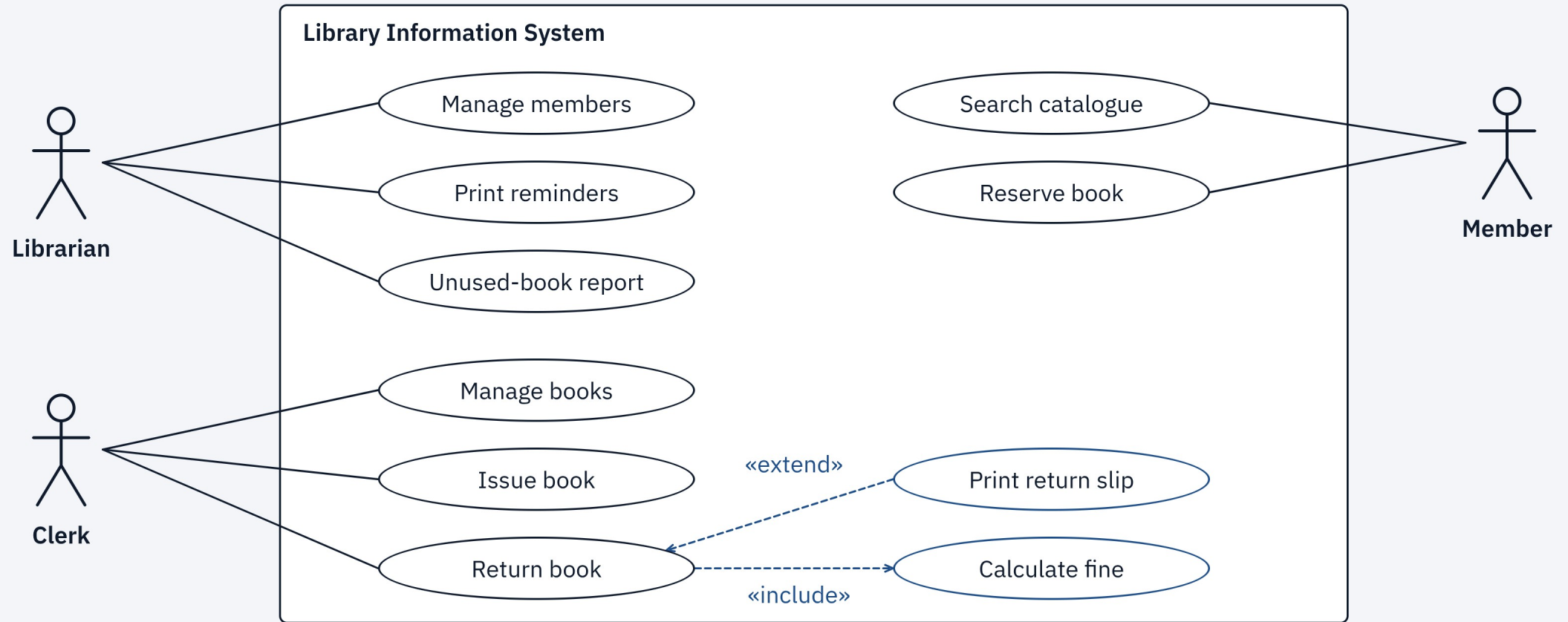
DFD level 1: catalogue, search, members, reports



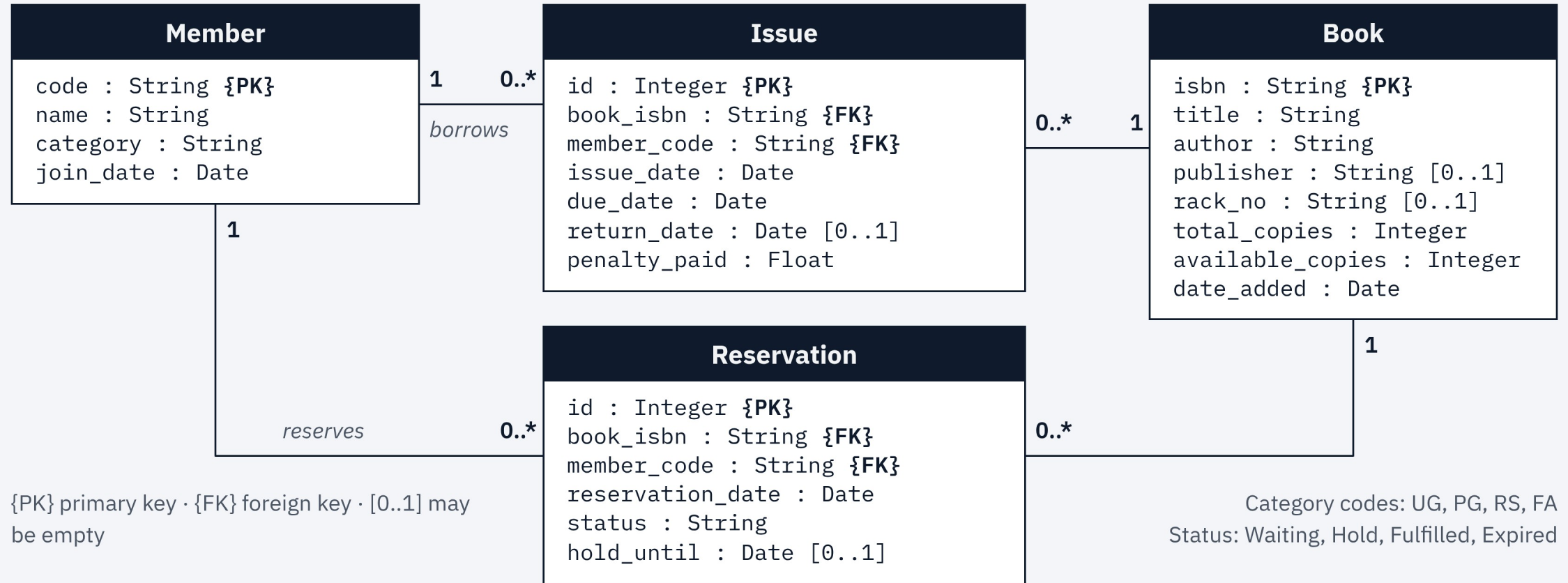
DFD level 1: issue, return, reserve



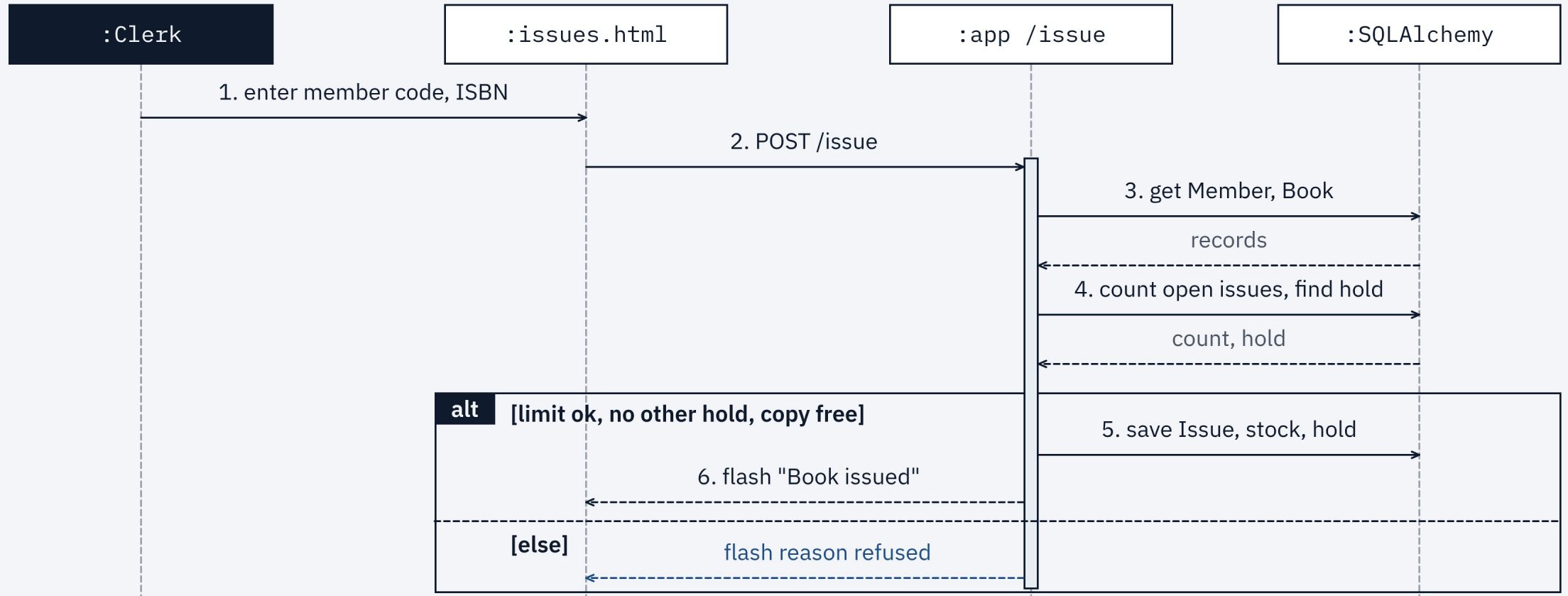
Use case diagram



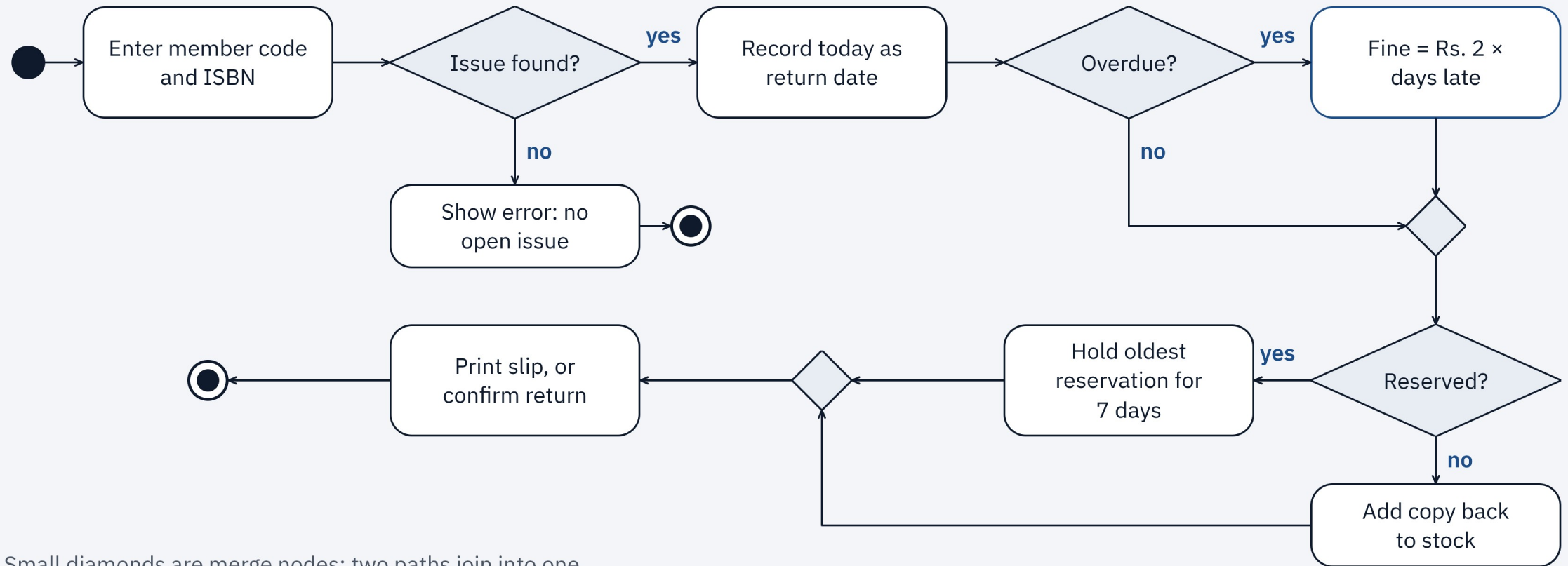
Class diagram



Sequence diagram: issuing a book

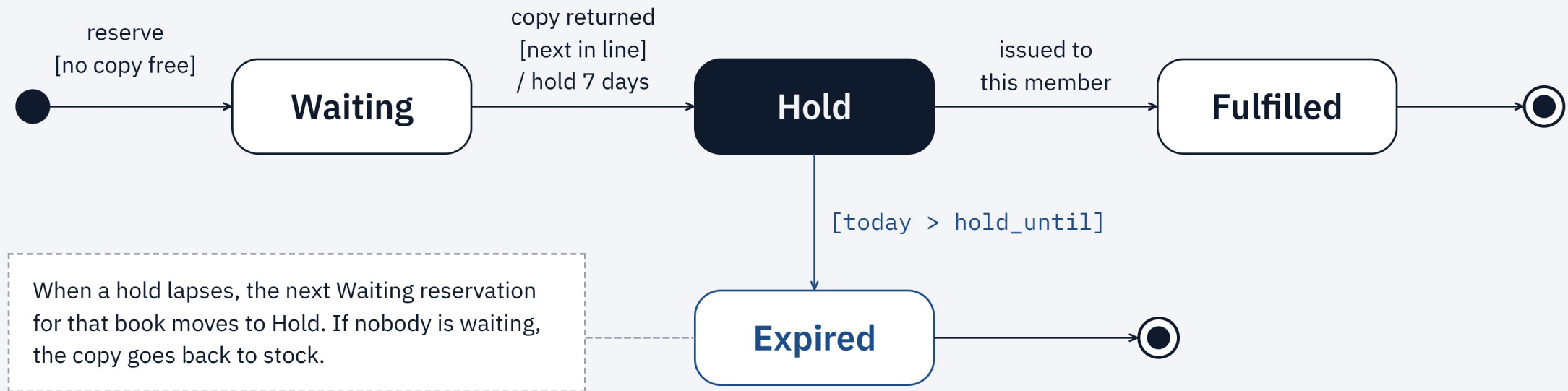


Activity diagram: returning a book



Small diamonds are merge nodes: two paths join into one.

State diagram: reservation lifecycle



API reference: sign-in and catalogue

Method	Endpoint	Role	Purpose
GET	/	Anyone	Login page
POST	/login	Anyone	Start a session
GET	/logout	Signed in	End the session
GET	/dashboard	Signed in	Menu for the role
GET	/books	Anyone	Catalogue
POST	/books/add	Clerk	Add a book
GET	/books/delete/<isbn>	Clerk	Remove a book
GET	/search?q=	Anyone	Search the catalogue

API reference: circulation and reports

Method	Endpoint	Role	Purpose
GET, POST	/members	Librarian	List or add members
GET	/members/delete/<code>	Librarian	Remove a member
GET, POST	/issue	Clerk	Issue a book
GET, POST	/return	Clerk	Return a book
POST	/reserve	Member	Reserve a book
GET	/reports/reminders	Librarian	Overdue reminders
GET	/reports/unused	Librarian	Unused-book report

Pages and how they differ

Page	Shows
login.html	Role picker; code field only for members
dashboard.html	A different menu for each role
books.html	Catalogue with add and delete
book_search.html	Results; Reserve for members when none free
members.html	Member list and create form
issues.html	One form shared by Issue and Return
print_return.html	Penalty bill and reservation slip
print_reminder.html	One overdue reminder per page
report_unused.html	Books not issued in five years
base.html	Shared navbar, messages and print styles

The app running on demo data

LIS Demo

Role: Clerk (Clerk)DashboardLogout

Print DocumentBack to Returns

PENALTY BILL

Member Code: P001

Book ISBN: B002 (Design Patterns)

Due Date: 2026-09-23

Return Date: 2026-09-28

Penalty Amount: Rs. 10.0

RESERVATION SLIP

The following reserved book is now available:

Member Code: R001

Book ISBN: B002 (Design Patterns)

Available Until: 2026-10-05

Please present this slip at the circulation desk within 7 days.

Clerk: late return, bill and hold slip

LIS Demo

Role: Member (R001)DashboardLogout

Library Catalog

DesignSearch

Results for "Design"

ISBN	Title	Author	Rack	Available	Action
B002	Design Patterns	GoF	R1	Out of stock	Reserve

Member: reserve a checked-out book

LIS Demo

Role: Librarian (Librarian)DashboardLogout

Dashboard - Welcome Librarian

MembersManage Members

ReportsOverdue RemindersUnused Books (5 Yrs)

Librarian: members and reports

Unit tests: 3 of 3 pass

Test	Checks	Result
test_due_date_calculation	UG due date is +30 days	Pass
test_penalty_calculation	5 days late costs Rs. 10	Pass
test_reservation_hold	Hold lasts 7 days	Pass

```
$ pytest tests/
```

```
3 passed in 0.24s
```

Backend test cases

Test case	Expected	Result
Issue beyond the category limit	Refused	Pass
Issue a book with no free copy	Refused	Pass
Issue a copy held for someone else	Refused	Pass
Return on the due date	No fine	Pass
Return five days late	Rs. 10 fine	Pass
Add a duplicate ISBN	Refused	Pass
Delete a book with copies on loan	Refused	Pass
Delete a member with unreturned books	Refused	Pass
Reserve the same book twice	Refused	Pass

Frontend test cases

Test case	Expected	Result
Submit the issue form empty	Browser blocks it	Pass
Enter 0 copies for a new book	Browser blocks it	Pass
Log in with an unknown code	Error message	Pass
Member opens a clerk page	Access denied	Pass
Click Delete on a book	Asks to confirm	Pass
Member finds an issued-out book	Reserve button	Pass
Print a penalty bill	No navbar or buttons	Pass
Search for ' OR 1=1 --	Treated as text	Pass

Integration tests: end-to-end scenarios

Scenario	Outcome	Result
Late return while a member waits	Bill and hold slip printed	Pass
Another member asks for that copy	Refused during the hold	Pass
Waiting member collects it	Issued; reservation fulfilled	Pass
Hold passes its end date	Next member moves to hold	Pass
Seed data, then test suite	3 passed	Pass

Future scope

- 🔒 Password login for staff
- ▶ Email reminders for overdue books
- 🔒 Online payment of fines
- 🔍 Barcode scanning at the counter

Thank you

